

NT-C-007 | Azure DevOps | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Cloud & DevOps
Target learners	Learners targeting Azure cloud operations and software-delivery automation.
Prerequisites	Basic networking, operating systems and Git concepts.
Approved duration	Approved working range: 16-20 weeks / 180-240 guided hours.
Final outcome	Plan, build and monitor a secure Azure delivery workflow using repositories, boards, pipelines, containers and infrastructure as code.
Assessment profile	Infrastructure/Security

Approved tools / platforms

Microsoft Azure, Azure Boards, Azure Repos, Azure Pipelines, Docker, ACR, AKS, Bicep/Terraform, Azure Monitor, Key Vault.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-007 | Azure DevOps | NT-C-007-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Identity, subscriptions, network, compute, storage; Git workflow, Azure Repos, pull requests
Completion evidence	Azure environment lab; Reviewed repository
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - Azure foundation

Objective: Identity, subscriptions, network, compute, storage

Observable evidence: Azure environment lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Identity	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when identity is appropriate; apply it in a Azure DevOps task; verify the result.	Azure environment lab
subscriptions	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when subscriptions is appropriate; apply it in a Azure DevOps task; verify the result.	Azure environment lab
network	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when network is appropriate; apply it in a Azure DevOps task; verify the result.	Azure environment lab
compute	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when compute is appropriate; apply it in a Azure DevOps task; verify the result.	Azure environment lab
storage	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when storage is appropriate; apply it in a Azure DevOps task; verify the result.	Azure environment lab

Module 2 - Source control

Objective: Git workflow, Azure Repos, pull requests

Observable evidence: Reviewed repository

Concept	Approved knowledge statement	Application behaviour	Evidence
Git workflow	Distributed version control for tracking and reviewing source changes. In Azure DevOps, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when git workflow is appropriate; apply it in a Azure DevOps task; verify the result.	Reviewed repository
Azure Repos	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when azure repos is appropriate; apply it in a Azure DevOps task; verify the result.	Reviewed repository
pull requests	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when pull requests is appropriate; apply it in a Azure DevOps task; verify the result.	Reviewed repository

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Azure environment lab; Reviewed repository, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-007 | Azure DevOps | NT-C-007-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Azure Boards, backlogs, sprints, traceability; YAML, agents, tests, artifacts
Completion evidence	Delivery board; Continuous integration pipeline
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Work planning

Objective: Azure Boards, backlogs, sprints, traceability

Observable evidence: Delivery board

Concept	Approved knowledge statement	Application behaviour	Evidence
Azure Boards	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when azure boards is appropriate; apply it in a Azure DevOps task; verify the result.	Delivery board
backlogs	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when backlogs is appropriate; apply it in a Azure DevOps task; verify the result.	Delivery board
sprints	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sprints is appropriate; apply it in a Azure DevOps task; verify the result.	Delivery board
traceability	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when traceability is appropriate; apply it in a Azure DevOps task; verify the result.	Delivery board

Module 4 - Build pipelines

Objective: YAML, agents, tests, artifacts

Observable evidence: Continuous integration pipeline

Concept	Approved knowledge statement	Application behaviour	Evidence
YAML	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when yaml is appropriate; apply it in a Azure DevOps task; verify the result.	Continuous integration pipeline
agents	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when agents is appropriate; apply it in a Azure DevOps task; verify the result.	Continuous integration pipeline
tests	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tests is appropriate; apply it in a Azure DevOps task; verify the result.	Continuous integration pipeline
artifacts	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when artifacts is appropriate; apply it in a Azure DevOps task; verify the result.	Continuous integration pipeline

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Delivery board; Continuous integration pipeline, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-007 | Azure DevOps | NT-C-007-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Environments, approvals, deployment strategies; Docker, ACR, AKS foundation
Completion evidence	Controlled deployment; Container deployment
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - Release pipelines

Objective: Environments, approvals, deployment strategies

Observable evidence: Controlled deployment

Concept	Approved knowledge statement	Application behaviour	Evidence
Environments	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when environments is appropriate; apply it in a Azure DevOps task; verify the result.	Controlled deployment
approvals	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when approvals is appropriate; apply it in a Azure DevOps task; verify the result.	Controlled deployment
deployment strategies	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deployment strategies is appropriate; apply it in a Azure DevOps task; verify the result.	Controlled deployment

Module 6 - Containers

Objective: Docker, ACR, AKS foundation

Observable evidence: Container deployment

Concept	Approved knowledge statement	Application behaviour	Evidence
Docker	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when docker is appropriate; apply it in a Azure DevOps task; verify the result.	Container deployment
ACR	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when acr is appropriate; apply it in a Azure DevOps task; verify the result.	Container deployment
AKS foundation	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when aks foundation is appropriate; apply it in a Azure DevOps task; verify the result.	Container deployment

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Controlled deployment; Container deployment, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-007 | Azure DevOps | NT-C-007-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Bicep/Terraform, variables, secrets; Monitor, logs, Key Vault, policy, capstone
Completion evidence	Repeatable Azure stack; Observed secure platform
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Infrastructure as code

Objective: Bicep/Terraform, variables, secrets

Observable evidence: Repeatable Azure stack

Concept	Approved knowledge statement	Application behaviour	Evidence
Bicep/Terraform	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when bicep/terraform is appropriate; apply it in a Azure DevOps task; verify the result.	Repeatable Azure stack
variables	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when variables is appropriate; apply it in a Azure DevOps task; verify the result.	Repeatable Azure stack
secrets	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when secrets is appropriate; apply it in a Azure DevOps task; verify the result.	Repeatable Azure stack

Module 8 - Operations & security

Objective: Monitor, logs, Key Vault, policy, capstone

Observable evidence: Observed secure platform

Concept	Approved knowledge statement	Application behaviour	Evidence
Monitor	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when monitor is appropriate; apply it in a Azure DevOps task; verify the result.	Observed secure platform
logs	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logs is appropriate; apply it in a Azure DevOps task; verify the result.	Observed secure platform
Key Vault	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when key vault is appropriate; apply it in a Azure DevOps task; verify the result.	Observed secure platform
policy	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when policy is appropriate; apply it in a Azure DevOps task; verify the result.	Observed secure platform
capstone	A defined competency within Azure DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when capstone is appropriate; apply it in a Azure DevOps task; verify the result.	Observed secure platform

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Repeatable Azure stack; Observed secure platform, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-007 | Azure DevOps | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-007-SK01	Azure foundation	Prerequisite foundation	Azure environment lab	Critical
NT-C-007-SK02	Source control	M01 plus earlier foundations	Reviewed repository	Critical
NT-C-007-SK03	Work planning	M02 plus earlier foundations	Delivery board	Supporting
NT-C-007-SK04	Build pipelines	M03 plus earlier foundations	Continuous integration pipeline	Supporting
NT-C-007-SK05	Release pipelines	M04 plus earlier foundations	Controlled deployment	Supporting
NT-C-007-SK06	Containers	M05 plus earlier foundations	Container deployment	Critical
NT-C-007-SK07	Infrastructure as code	M06 plus earlier foundations	Repeatable Azure stack	Supporting
NT-C-007-SK08	Operations & security	M07 plus earlier foundations	Observed secure platform	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-007 | Azure DevOps | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-007-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Azure foundation and Source control?	The learner should provide Azure environment lab; Reviewed repository and explain how the work applies Identity, subscriptions, network, compute, storage; Git workflow, Azure Repos, pull requests. Recall alone is insufficient.
NT-C-007-S01-EX02	Scenario/remediation	A learner reports completing Source control but cannot produce Reviewed repository. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-007-S01, request Reviewed repository, and allow a new equivalent test only after remediation.
NT-C-007-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Work planning and Build pipelines?	The learner should provide Delivery board; Continuous integration pipeline and explain how the work applies Azure Boards, backlogs, sprints, traceability; YAML, agents, tests, artifacts. Recall alone is insufficient.
NT-C-007-S02-EX02	Scenario/remediation	A learner reports completing Build pipelines but cannot produce Continuous integration pipeline. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-007-S02, request Continuous integration pipeline, and allow a new equivalent test only after remediation.
NT-C-007-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Release pipelines and Containers?	The learner should provide Controlled deployment; Container deployment and explain how the work applies Environments, approvals, deployment strategies; Docker, ACR, AKS foundation. Recall alone is insufficient.
NT-C-007-S03-EX02	Scenario/remediation	A learner reports completing Containers but cannot produce Container deployment. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-007-S03, request Container deployment, and allow a new equivalent test only after remediation.
NT-C-007-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Infrastructure as code and Operations & security?	The learner should provide Repeatable Azure stack; Observed secure platform and explain how the work applies Bicep/Terraform, variables, secrets; Monitor, logs, Key Vault, policy, capstone. Recall alone is insufficient.
NT-C-007-S04-EX02	Scenario/remediation	A learner reports completing Operations & security but cannot produce Observed secure platform. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-007-S04, request Observed secure platform, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-007 | Azure DevOps | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Repository-to-build workflow	Repos, pull requests, pipeline	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Container deployment	Docker, ACR, Azure target	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Secure Azure CI/CD platform	IaC, approvals, monitoring	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-007 | Azure DevOps | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Azure DevOps Engineer - Trainee	Repos 25% [critical]; pipelines 25% [critical]; IaC 25%; monitoring 25%	Automated Azure project	If a critical skill is below 60 or automated azure project is missing, return Improve First or Needs Evidence with the linked module/project.
2	Cloud Support Associate	Azure core 34% [critical]; identity 33% [critical]; networking 33%	Operations lab portfolio	If a critical skill is below 60 or operations lab portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
3	Build & Release Engineer - Junior	YAML 34% [critical]; artifacts 33% [critical]; environments 33%	Controlled pipeline evidence	If a critical skill is below 60 or controlled pipeline evidence is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-007 | Azure DevOps | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Azure DevOps for?	Learners targeting Azure cloud operations and software-delivery automation.
What should I know before starting?	Basic networking, operating systems and Git concepts.
How long is the approved learning journey?	Approved working range: 16-20 weeks / 180-240 guided hours.
What will I be able to demonstrate?	Plan, build and monitor a secure Azure delivery workflow using repositories, boards, pipelines, containers and infrastructure as code.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Azure DevOps knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-007; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```